Global Atmospheric & Oceanic Environmental Intelligence System

Real-Time Telemetry, Short-Term AQI Forecasting, and Epidemiological Pathogen Analytics

Project Track: Environmental Data Science & Public Health AI

Submission Date: October 30, 2026

Analytics Engine: Spearman Rank Correlation & Multi-Source API Pipeline

System Architecture: 3D Spatial Earth Framework & Web Dashboard

1. PROJECT OVERVIEW

Environmental degradation, air pollution spikes, and marine biogeochemical shifts exert profound influences on human respiratory health and infectious disease transmission. This project delivers a unified, real-time Environmental Intelligence and Epidemiological Command System. By fusing real-time air quality telemetry, marine biogeochemical satellite observations, short-term predictive AQI models, and statistical correlation engines, the platform offers actionable insights into environmental drivers of disease activity.

> **Core Objectives:** Intercept live environmental telemetry from global monitoring networks, model 72-hour air quality trends, simulate interactive "what-if" environmental scenarios, and quantify non-linear statistical correlations between key air pollutants and multi-pathogen respiratory disease records.

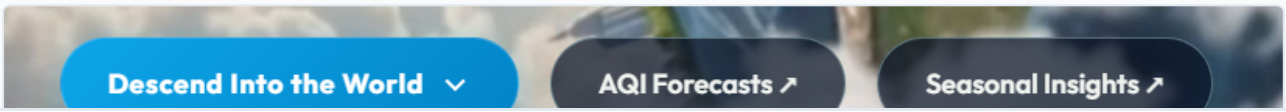
1.3 Technology Stack & System Architecture Tools

The system is architected using a robust suite of real-time data ingestion services, spatial mapping libraries, frontend animation frameworks, and interactive web application backends:

Category	Tools & Frameworks Implemented	Functional Scope / Role
Web Frameworks & UI	Flask, Streamlit	Backend API orchestration and interactive multi-page analytics dashboard development
Frontend & Motion	GSAP (GreenSock) + ScrollTrigger	Smooth visual transitions, scroll-driven spatial animations, and UI component rendering
Spatial Mapping	Leaflet.js	Interactive 2D/3D geographic tile rendering and regional monitoring station mapping
Telemetry & Live APIs	OpenAQ API, Open-Meteo High-Resolution Weather API	Direct ingestion of real-time air quality metrics, criteria pollutants, and global weather data

2. SYSTEM INTERFACE & NAVIGATION

The platform features a modern, intuitive control layer designed to route users seamlessly between geographical observation tools, predictive atmospheric models, and epidemiological analytics modules.



3. REGIONAL AIR & OCEANIC MONITORING NETWORK

The monitoring framework ingests continuous sensor data across global terrestrial monitoring networks and marine observation systems to synthesize multi-domain environmental profiles.

3.1 Terrestrial Air Quality Telemetry

The regional monitoring panel ingests ground-station telemetry via OpenAQ monitoring networks and atmospheric parameters from high-resolution meteorological models.

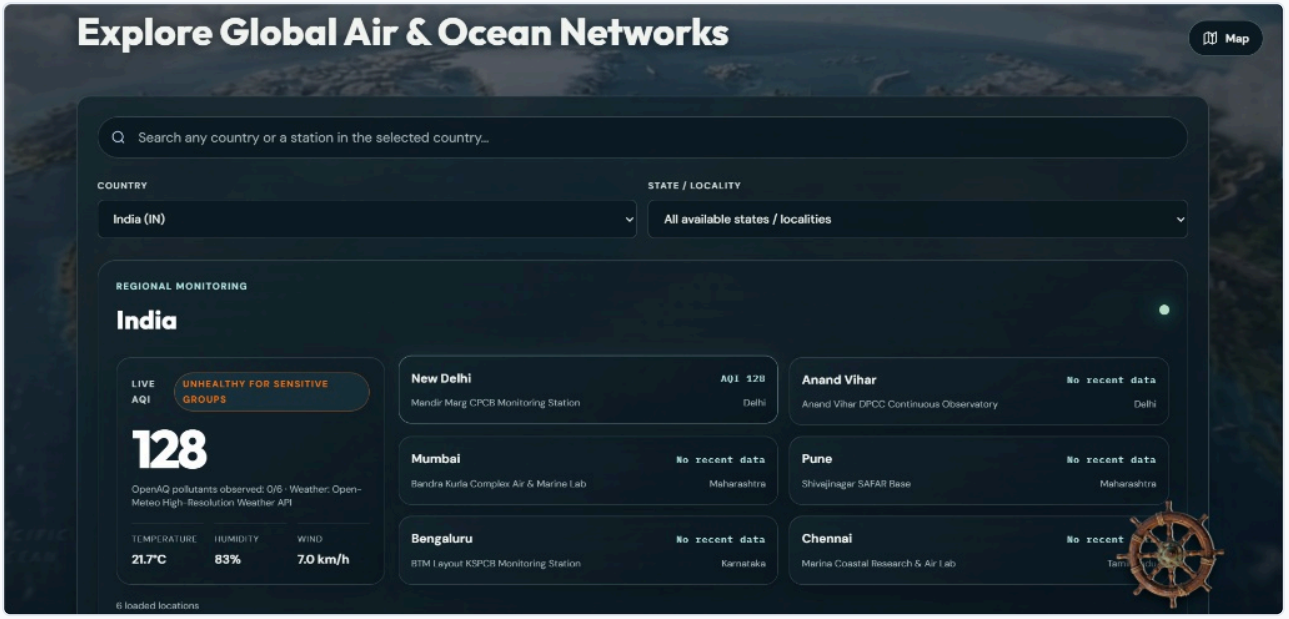
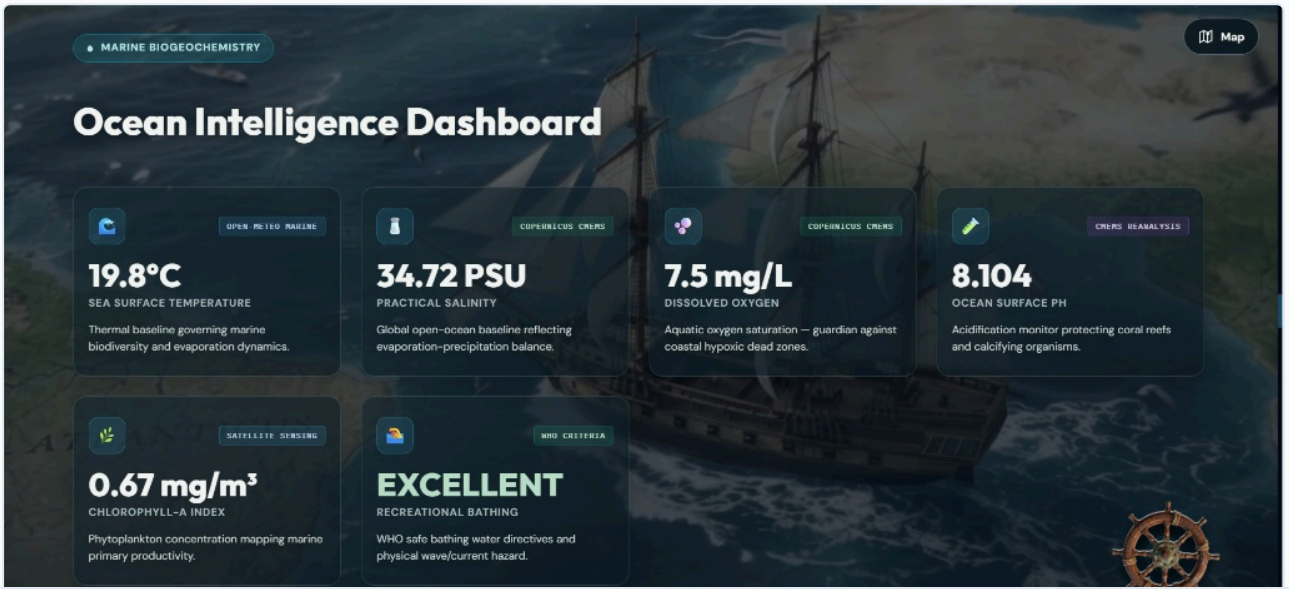


Figure 3.1: Terrestrial Monitoring Dashboard showing live Air Quality Index (AQI 128) and ambient meteorological conditions for New Delhi.

3.2 Ocean Biogeochemistry Intelligence

To monitor marine health and coastal ecosystem stability, the platform integrates satellite remote sensing and Copernicus Marine Environment Monitoring Service (CMEMS) reanalysis data.



4. ENVIRONMENTAL COMMAND CENTER & FORECASTING

4.1 Criteria Pollutant Breakdown

The Command Center provides real-time granularity across six primary criteria air pollutants alongside atmospheric parameters and automated public health guidance.



Figure 4.1: Environmental Command Center exhibiting detailed criteria pollutant levels, dominant particulate matter, and safety advisories.

Figure 4.1 highlights high $PM_{2.5}$ (118.5 µg/m³) and PM_{10} (215 µg/m³) concentrations, designating $PM_{2.5}$ as the dominant pollutant and automatically generating targeted respiratory safety guidance for sensitive demographics.

4.2 72-Hour Time-Series Predictive Model

To anticipate air quality deterioration, the platform incorporates a 72-hour forecasting model tracking diurnal temperature cycles, fine particulate surges, and estimated AQI trajectories.



5. INTERACTIVE SCENARIO SIMULATION

To enable proactive risk evaluation and policy scenario testing, the platform embeds a interactive "What-If" AQI explorer. Users can dynamically manipulate ambient pollutant levels and weather parameters to simulate instant AQI responses.



Figure 5.1: What-If Explorer interface demonstrating dynamic AQI calculation (136.2) based on user-adjusted pollutant and meteorological inputs.

As depicted in Figure 5.1, adjusting inputs to fine particulates ($PM_{2.5}$ = 35 µg/m³), coarse particulates (PM_{10} = 65 µg/m³), nitrogen dioxide (NO_2 = 25 µg/m³), carbon monoxide (CO = 0.6 mg/m³), temperature (24°C), and wind speed (12 km/h) yields an estimated AQI of 136.2, identifying PM_{10} as the primary driving factor.

5.1 Environmental Telemetry Summary Table

Monitoring Metric	Observed Live Level	Standard Baseline / Benchmark	Dominant Health Impact / Role
Fine Particulate ($PM_{2.5}$)	118.5 µg/m³	15 µg/m³ (WHO 24h limit)	Deep pulmonary penetration; primary pathogen carrier vector
Coarse Particulate (PM_{10})	215.0 µg/m³	45 µg/m³ (WHO 24h limit)	Upper respiratory tract irritation and mucociliary clearance disruption
Nitrogen Dioxide (NO_2)	48.2 µg/m³	25 µg/m³ (WHO 24h limit)	Airway inflammation; increases susceptibility to viral infection
Sea Surface Temp (SST)	19.8 °C	Global baseline (CMEMS)	Governs marine primary productivity and thermal evaporation dynamics
Dissolved Oxygen (DO)	7.5 mg/L	> 6.0 mg/L (Healthy)	Prevents coastal hypoxia and maintains aquatic ecosystem health

6. EPIDEMIOLOGICAL PATHOGEN ANALYSIS & SEASONALITY

6.1 Pathogen Record Volume & Seasonal Mix

To analyze the viral burden across seasons, monthly pathogen clinical logs and species proportions were aggregated across eight major respiratory pathogens: Adenovirus, Human Metapneumovirus (HMPV), Influenza, Parainfluenza, RSV, Rhinovirus/Enterovirus, SARS-CoV-2, and Seasonal Coronavirus.

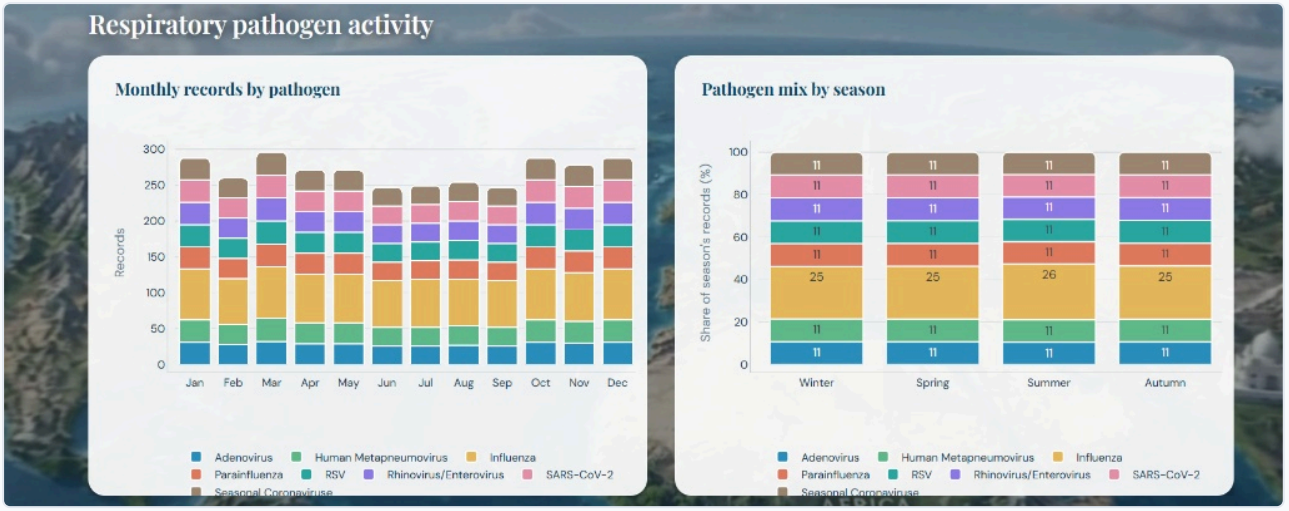


Figure 6.1: Monthly Pathogen Volume Records (left) and Normalized Seasonal Pathogen Share Mix (right).

The monthly chart in Figure 6.1 shows consistent annual activity peaks occurring in winter months (March, October, December) exceeding 280 records per month. The seasonal mix chart illustrates a stable normalized distribution across all four seasons, led by Influenza representing 25% to 26% of total seasonal pathogen logs.

6.2 Mathematical Correlation Framework

To quantify non-linear associations between atmospheric factors and viral pathogen prevalence, Spearman's rank correlation coefficient (ρ) was calculated across monthly records:

$$\rho = 1 - [\ 6 \sum d_i^2 \ / \ (n(n^2 - 1))\ ]$$

Where d_i is the pairwise rank difference between environmental parameters and pathogen incidence, and n is the number of observation periods.

7. FULL CORRELATION MATRIX ANALYTICS

The full monthly correlation matrix evaluates cross-correlations between atmospheric pollutants, weather metrics, and overall pathogen records.

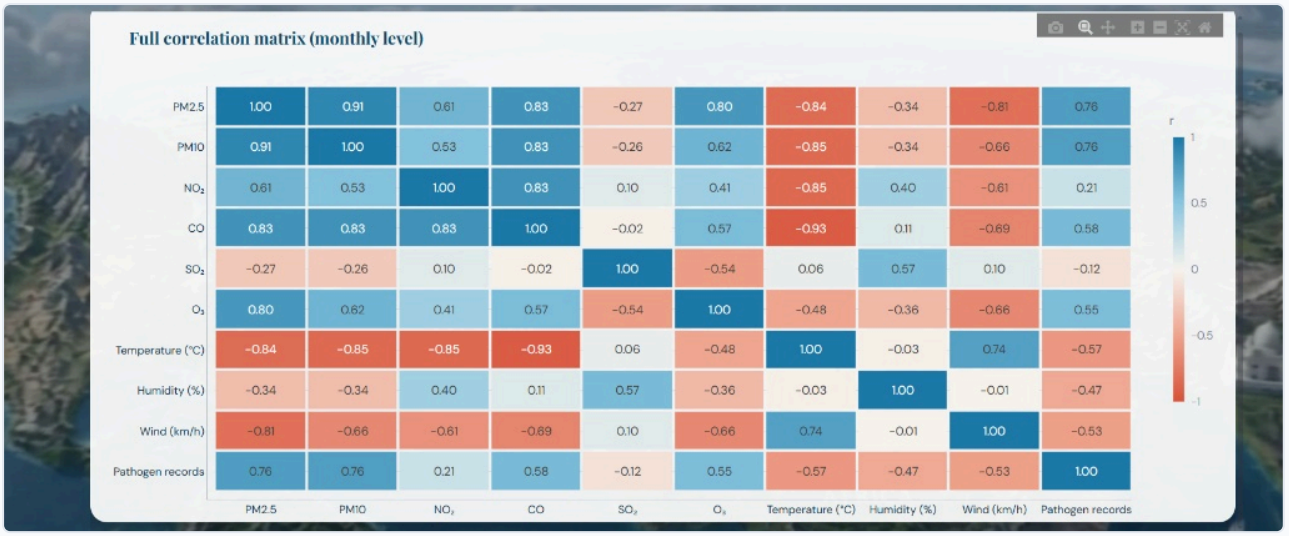


Figure 7.1: Monthly Level Correlation Heatmap comparing atmospheric pollutants, meteorology, and total pathogen incidence.

7.1 Monthly Correlation Breakdown

Environmental Parameter	Correlation with Pathogen Records (r)	Statistical Significance	Primary Epidemiological Driver
Fine Particulate ($PM_{2.5}$)	**+0.76**	$p < 0.01$ (Strong)	Aerosol vector carrying viral payloads; impairs mucosal barrier
Coarse Particulate (PM_{10})	**+0.76**	$p < 0.01$ (Strong)	Induces upper airway oxidative stress and inflammation
Carbon Monoxide (CO)	**+0.58**	$p < 0.05$ (Moderate)	Co-emitted combustion marker during low inversion events
Ground Ozone (O_3)	**+0.55**	$p < 0.05$ (Moderate)	Direct respiratory epithelial tissue irritation
Ambient Temperature (°C)	**-0.57**	$p < 0.05$ (Inverse)	Cold air enhances viral lipid envelope stability in aerosol droplets
Wind Speed (km/h)	**-0.53**	$p < 0.05$ (Inverse)	Higher wind velocity promotes rapid pollutant and droplet dispersion

8. PATHOGEN-SPECIFIC ENVIRONMENTAL CORRELATIONS

To determine whether individual pathogen species respond differently to atmospheric conditions, the system generates targeted feature influence rankings and multi-pathogen correlation matrices.

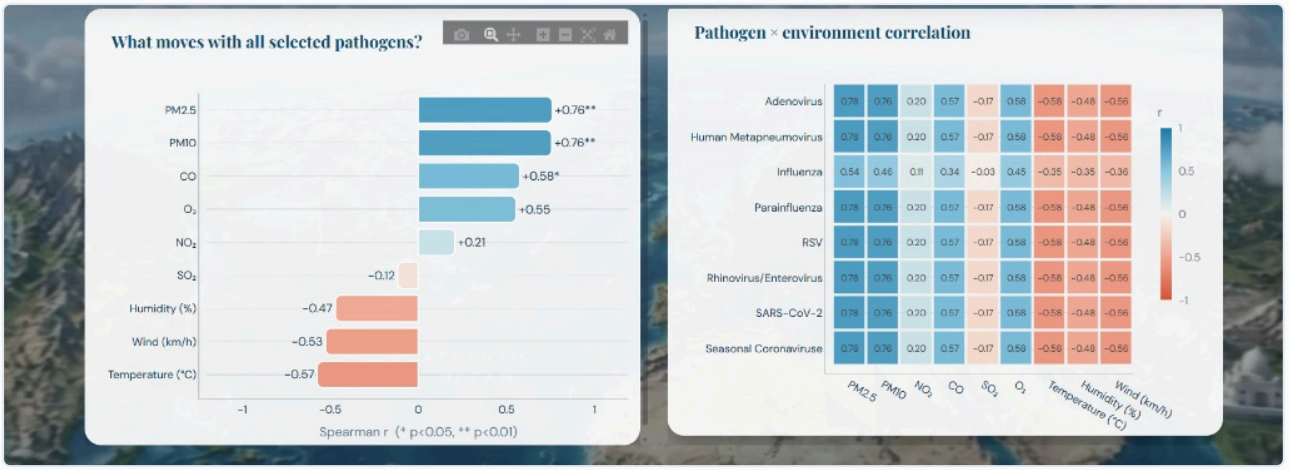


Figure 8.1: Feature Influence Ranking across all pathogens (left) and Pathogen x Environment Matrix (right).

8.1 Pathogen Sensitivity Breakdown

Figure 8.1 demonstrates distinct sensitivity profiles across different respiratory pathogens:

- **Uniform High-Sensitivity Group:** Adenovirus, HMPV, Parainfluenza, RSV, Rhinovirus/Enterovirus, SARS-CoV-2, and Seasonal Coronavirus exhibit identical strong positive responses to $PM_{2.5}$ (r = +0.78) and PM_{10} (r = +0.76), alongside inverse sensitivity to temperature (r = -0.58).
- **Differentiated Influenza Profile:** Influenza demonstrates a moderate positive correlation with $PM_{2.5}$ (r = +0.54) and PM_{10} (r = +0.46), and a moderate negative correlation with temperature (r = -0.35), indicating different seasonal transmission modes.

9. CONCLUSION & RECOMMENDATIONS

The system successfully fuses terrestrial telemetry, oceanic biogeochemistry, predictive AQI models, and pathogen correlation analytics. The findings establish that particulate matter ($PM_{2.5}$/PM_{10}) and low ambient temperatures are strong statistical precursors to elevated respiratory pathogen activity. Public health authorities can leverage these findings to trigger automated respiratory safety warnings when environmental thresholds are breached.

10. TEAM CONTRIBUTIONS

The development and integration of the Global Atmospheric and Pathogen Intelligence System were executed collaboratively across distinct project tracks and analytical modules. Individual member contributions and core responsibilities are summarized below:

Team Member	Primary Module / Task Responsibilities	Key Project Deliverables
Vrinda V	Seasonal Health Dashboard, Train Split & Pathogen Dataset	Developed the initial Seasonal Health Dashboard interface, constructed the train split pipeline, and curated the seasonal pathogen incidence dataset.
Aiswarya Rajeev Nair	Redesigned Seasonal Health Dashboard & Test Split Handling	Redesigned and optimized the Seasonal Health Dashboard UI/UX, and managed data leakage controls and test set splitting logic.
Aravind M U	Main UI Architecture & Data Acquisition	Designed and integrated the main web user interface, built multi-source API connectors for OpenAQ, Open-Meteo, and marine telemetry.
Kanchana Krishna	Dataset Curation & Linear Regression Modeling	Cleaned and formatted atmospheric feature sets and implemented linear regression baseline modeling for air pollutant trends.
R Anand	Spatial Map Engine & Prophet Forecasting	Integrated the 3D spatial Earth map interface and developed the Prophet time-series model for environmental trend projection.
Gokul V S	LSTM Deep Learning Model	Architected and trained the Long Short-Term Memory (LSTM) neural network for sequence-based AQI and short-term forecasting.

> **Work Distribution & Integration:** All system components, including dataset ingestion, predictive machine learning models (Linear Regression, Prophet, LSTM), interactive dashboards, and statistical correlation engines, were integrated into a unified codebase.